
EXPERIMENTAL ANALYSIS OF THE EFFECTS OF SELECTED WATER QUALITY PARAMETERS' ON SOIL INFILTRATION DYNAMICS UNDER CONTROLLED TEMPERATURES

Kalu P.N¹, Egwuonwu C.C², Nwakuba N.R²

¹Department of Agricultural and Bio-Environmental Engineering, Federal Polytechnic
Nekede,

Owerri – Imo State Nigeria

²Department of Agricultural and BioSystem Engineering, Federal University of Technology
Owerri –Imo State Nigeria

Corresponding author's email: nzekalu@yahoo.com, +2348030960175

ABSTRACT

The movement of water into soil is a critical process influencing irrigation efficiency, infiltration processes, groundwater recharge, soil moisture availability, and sustainable agricultural production. While soil characteristics have traditionally been regarded as the principal factors controlling infiltration, the influence of the physical quality of infiltrating water has received comparatively less attention. This study experimentally analyzed the effects of viscosity, density, and turbidity, on soil infiltration dynamics under controlled temperature conditions of 20°C, 25°C, 30°C, and 35°C. The study was conducted on an agricultural soil. The land slope/topography was determined as 1.7%, showing the land area to be relatively flat with a climatic condition of high rainfall and atmospheric moisture. Soil samples were collected from three locations within the experimental plot and subjected to laboratory analysis with the following results obtained: particle size distribution as 81% sand and 18% silt, proving the area to be loam-sand dominated, average moisture contents of 4.50%, 5.21%, and 4.90% for Locations A, B, and C, respectively, showing that the area has low moisture content. Hydraulic conductivity values of 6.505×10^{-3} cm/s, 6.513×10^{-3} cm/s, and 6.505×10^{-3} cm/s respectively were obtained, indicating relatively high permeability. Infiltration rates gradually decreased with time from 15.6 cm/h to 7.8 cm/h, 25.2 cm/h to 6.6 cm/h, and 20.4 cm/h to 7.8 cm/h at Locations A, B, and C, respectively, which shows that the area has good infiltration characteristics. Water samples were subjected to water test. The results showed that viscosity decreased with increasing temperature. The mean viscosity decreased from 5.072 mPa·s at 20°C to 4.298 mPa·s at 35°C, representing an overall reduction of approximately 15.3%. Two-way ANOVA showed that temperature significantly affected viscosity ($F(3,33) = 36.24$, $p < 0.001$), while water source had no significant effect ($F(11,33) = 1.31$, $p = 0.261$). Density also decreased progressively with increasing temperature, while both temperature ($F(3,33) = 47.47$, $p < 0.001$) and water source ($F(11,33) = 24.48$, $p < 0.001$) significantly affected density, with the statistics explaining 92.6% of the observed variation ($R^2 = 0.926$). Turbidity showed considerable variation among water sources, with borehole and rainfall water recording relatively low values, while surface water, ponds, rivers, and floodwater recorded considerably higher values, with some exceeding 1400 NTU. Water source was the dominant factor influencing turbidity ($F(11,33) = 4356.1$, $p < 0.001$), while temperature had a statistically significant but relatively small

effect ($F(3,33) = 17.27, p < 0.001$). Mean turbidity increased from 270.3 NTU at 20°C to 297.9 NTU at 35°C. Among the parameters investigated, viscosity was the most temperature-sensitive, followed by density; however, there was an exception in the turbidity parameter where the turbidity value was observed to be increasing with an increase in temperature, thereby influencing infiltration primarily through suspended sediment deposition and pore blockage. The study therefore establishes that cleaner water sources with low turbidity are more suitable for maintaining soil infiltration, while temperature should be treated as an essential interacting variable in the assessment of water quality effects on soil infiltration dynamics.

Keywords: Infiltration, Water-Quality, Viscosity, Density, Turbidity.

I. Introduction.

Water infiltration into soil is one of the most important processes in agricultural water management, irrigation engineering, groundwater recharge, and environmental sustainability. Infiltration refers to the downward movement of water from the soil surface into the soil profile where it becomes available for plant uptake, groundwater recharge, and maintenance of soil moisture (Michael & Ojha, 2005).

From the agricultural definition, according to Schwab, Fangmeier, Elliot, & Frevert (2021), infiltration is said to be the process by which precipitation moves downward through the surface of the soil to replenish soil moisture, recharge the aquifer and ultimately support stream flow during dry periods, as well as support plant growth.

Infiltration process of fluids (water) always comes into action during precipitation forms, e.g. dew, moist, rainfall, reaching the surface of the earth and also during irrigation processes (controlled artificial application of fluids (water) to the soil surface, principally for the purpose of supplying soil moisture essential for plant growth applied on the soil surface (Michael & Ojha, 2018). However, not all water applied or reaching the surface of the earth infiltrates easily or freely into the soil. Some water is ponded on the soil surface, thereby resisting free downward movement, while some gradually move downwards into the soil, causing a delay in the rate or time of infiltration (Ogbu, Nwachukwu, Orakwe, & Chukwuma, 2015).

However, research has shown that the cause of this resistance/delay or gradual flow in the downward movement of various forms of fluids (water) from the surface of the earth into the soil, is found to be influenced by several factors, including soil texture, soil structure, porosity, antecedent moisture content, organic matter content, and the physical properties of the infiltrating fluid (water quality), either used for irrigation on that soil. Among the fluid properties that influence infiltration are viscosity, turbidity, and density (Arora, 2009; Aboukhalesl, Nofal, & Hisham, 2020).

Water quality refers to the chemical, physical, and biological characteristics of water based on the standard of its usage, (Cordy, 2001). Water quality is a multidimensional concept that describes the physical, chemical, and biological characteristics of water and determines its suitability for a specific purpose such as drinking, irrigation, industrial processes, aquaculture, or environmental conservation.

Among the numerous water quality parameters, the physical properties of water (viscosity, density, and turbidity) are particularly significant because they directly influence the hydraulic behavior of water, especially during infiltration, seepage, groundwater recharge,

and irrigation. Consequently, understanding the relationship between water quality and these three physical properties as well as their influence is fundamental in irrigation engineering, hydrology, soil science, and environmental management (WHO, 2022; APHA, AWWA, & WEF, 2023). The physical quality of water influences its movement through soil pores, interaction with soil particles, and ability to transport dissolved and suspended materials. Variations in these water qualities (viscosity, density, and turbidity) often alter the resistance of water to flow, the gravitational forces acting on water within the soil profile, and the extent to which suspended particles obstruct soil pore spaces. These changes ultimately affect soil infiltration rate, hydraulic conductivity, soil aeration, groundwater recharge, and crop productivity (Hillel, 1998; Dingman, 2015).

Viscosity, density, and turbidity are presumed to be some of the factors affecting infiltration during the irrigation period on the agricultural soil. Most infiltration studies focus mainly on soil characteristics as factors that influence the downward movement of water into the soil, giving limited attention to the influence of these selected water quality parameters (viscosity, density, and turbidity parameters). Usually, Variations in viscosity, turbidity, and density may significantly alter infiltration behaviour, affecting soil moisture replenishment, groundwater recharge, irrigation efficiency, and crop productivity. Therefore, there is a need to investigate how these water quality parameters influence infiltration rates under varying temperature conditions to provide useful information for irrigation planning and water resource management.

Objective of the Study

The main objective of this study is to experimentally analyze the effects of selected water quality parameters (viscosity, turbidity, and density) on soil infiltration dynamics under controlled temperature conditions of 20, 25, 30°C, and 35°C.

Other specific objectives includes to:

- i. analyze the viscosity, density and turbidity of the different water samples at 20, 25, 30, and 35°C and their effects on soil infiltration.
- ii. determine the infiltration characteristics of different water samples at different temperatures.
- iii. statistically analyze the temperature relationship between the selected water quality parameters and soil infiltration.
- iv. specify the water quality parameter with the greatest influence on soil infiltration under the experimental conditions.

II. Literature Review

Water quality refers to the physical, chemical, and biological characteristics of water that determine its suitability for a particular purpose, such as drinking, irrigation, industrial use, recreation, or environmental sustainability. The quality of irrigation water is of particular importance because it directly affects soil physical properties, infiltration characteristics, nutrient availability, crop performance, and long-term soil productivity. Water containing excessive suspended solids, dissolved salts, organic matter, or contaminants may gradually alter soil structure, reduce hydraulic conductivity, clog soil pores, and decrease infiltration rates (WHO, 2022; APHA, AWWA, & WEF, 2023).

Physical water quality refers to the observable and measurable physical characteristics of water which determine its appearance, movement, and hydraulic behaviour. The major physical parameters include temperature, colour, odour, taste, turbidity, total suspended

solids (TSS), total dissolved solids (TDS), electrical conductivity (EC), density, and viscosity.

Viscosity is one of the most important physical properties of water affecting fluid flow. In fluid mechanics, viscosity represents the friction that exists between adjacent layers of a moving fluid. It is defined as the internal resistance of a fluid to deformation or flow under an applied force. Fluids with high viscosity resist movement, whereas fluids with low viscosity flow more readily (Potter, Wiggert and Ramadan, 2025).

Temperature has a significant effect on the viscosity of fluids (water). As water temperature increases, the cohesive forces between molecules reduce, resulting in reduced viscosity. Thus, at high temperatures water loses its viscosity, while at low temperatures its viscosity increases. The viscosity of water varies with temperature. The values presented by Rumble and John (2018) show that dynamic and kinematic viscosity decrease as temperature increases. For example, the dynamic viscosity of water decreases from $1.002 \times 10^{-3} \text{ Ns/m}^2$ at 20°C to $0.798 \times 10^{-3} \text{ Ns/m}^2$ at 30°C and $0.653 \times 10^{-3} \text{ Ns/m}^2$ at 40°C . Thus, water at low temperatures has higher viscosity, while water at higher temperatures has lower viscosity.

Water, which is the material used in this work, falls under the Newtonian fluid classification, but may change with slight changes in environmental conditions such as temperature and pressure. The viscosity coefficient of a liquid decreases with increasing temperature, while the coefficient of viscosity of gases increases with increasing temperature.

Fluid viscosity controls the resistance to flow within interconnected pore spaces; consequently, an increase in viscosity decreases hydraulic conductivity (Assouline & Or, 2013; Wang, Ma, & Zhu, 2022). As viscosity decreases, hydraulic resistance is reduced, resulting in increased infiltration rates and more rapid water movement under both saturated and unsaturated conditions (Kirkham, 2014).

Density is one of the fundamental physical properties of fluids and is defined as the mass of a substance per unit volume. It is an important parameter in fluid mechanics, hydrology, irrigation engineering, and soil-water movement because it influences the gravitational forces acting on water during flow and infiltration processes. Under standard atmospheric conditions, pure water has a density of approximately 998 kg/m^3 at 20°C . As temperature increases, water molecules become more energetic and move farther apart, resulting in a slight decrease in density. Density plays a significant role in determining the energy state and movement of water within soil pores. Water infiltration is driven primarily by gravitational forces and capillary action.

Turbidity is also one of the most important physical indicators of water quality. Turbid water usually appears cloudy, milky or otherwise coloured, thereby affecting the physical look of the water. Turbidity is also a measure of the degree to which water loses its transparency because of the presence of suspended and colloidal particles that scatter and absorb light passing through the water column (Lipps, Braun-Howland, & Baxter, 2023). The major source of turbidity is total suspended solids (TSS). These suspended solids come from two sources: one is a natural source, and the other is an anthropogenic source. The natural sources are sediments such as silts, clay, organic materials, algae, dead organisms, etc., while anthropogenic sources are wastewater (sewage), agricultural runoff and erosion from rainfall. Turbidity is expressed in Nephelometric Turbidity Units (NTU) and is measured using instruments known as nephelometers or turbidimeters.

In irrigation systems, water turbidity directly influences soil infiltration because suspended sediments are deposited on the soil surface and within soil pore spaces during irrigation. Over time, these fine particles form a thin, low-permeability layer (commonly referred to as a surface seal or clogging layer), which disrupts pore continuity, reduces saturated hydraulic

conductivity, and significantly decreases infiltration rates (Kang et al., 2023; Xiao et al., 2023).

Infiltration is the process by which precipitation or any form of water moves downward through the surface of the soil into the soil in order to replenish soil moisture, recharge aquifer, and ultimately support stream flow during dry periods. The maximum rate at which the water enters the soil in a given time is known as the infiltration capacity while a measure of the rate at which the soil is able to absorb the water is known as the infiltration rate and it is measured in millimeter per hour (mm/hr) or inches per hour (Kirkham, 2014). The rate of infiltration is usually measured using an infiltrometer (Nimmo & Shilliti, 2023).

The infiltration rate of water into soil is affected by soil characteristics, storage capacity, soil texture, vegetation type and cover, moisture content, soil temperature, rainfall intensity, and frost (Huffman et al., 2013).

III. Materials and Methods

Study Area

The study was conducted at the Teaching and Research Farm of the School of Agriculture and Agricultural Technology, Federal Polytechnic Nekede, located in Nekede, Owerri West Local Government Area of Imo State, southeastern Nigeria (Federal Polytechnic Nekede, n.d.). The institution lies approximately between Latitude 5°27'–5°30' N and Longitude 6°58'–7°02' E, at an elevation of about 75–90 m above mean sea level. The study area experiences a humid tropical climate characterized by two distinct seasons: a rainy season extending from April to October and a dry season from November to March. Mean annual rainfall ranges between 2,000 and 2,500 mm, while average temperatures vary from 25°C to 35°C. Relative humidity ranges from about 80% during the dry season to over 100% during the rainy season. The soils are predominantly sandy loam to loamy sand, well-drained, porous, and characterized by moderate to high infiltration capacity, making them highly suitable for studies involving soil-water movement.

The map of the study location showing the bounded communities and the geographical features, is shown below.

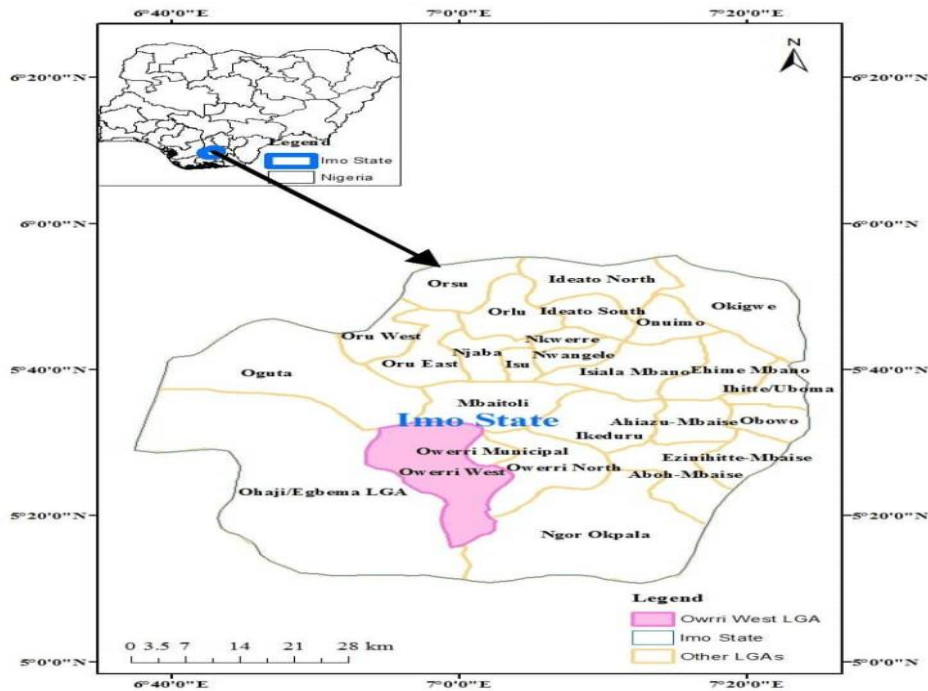


Figure 1: Map of Imo State indicating Owerri West Local Government Area Source: (RECTAS, 2013)

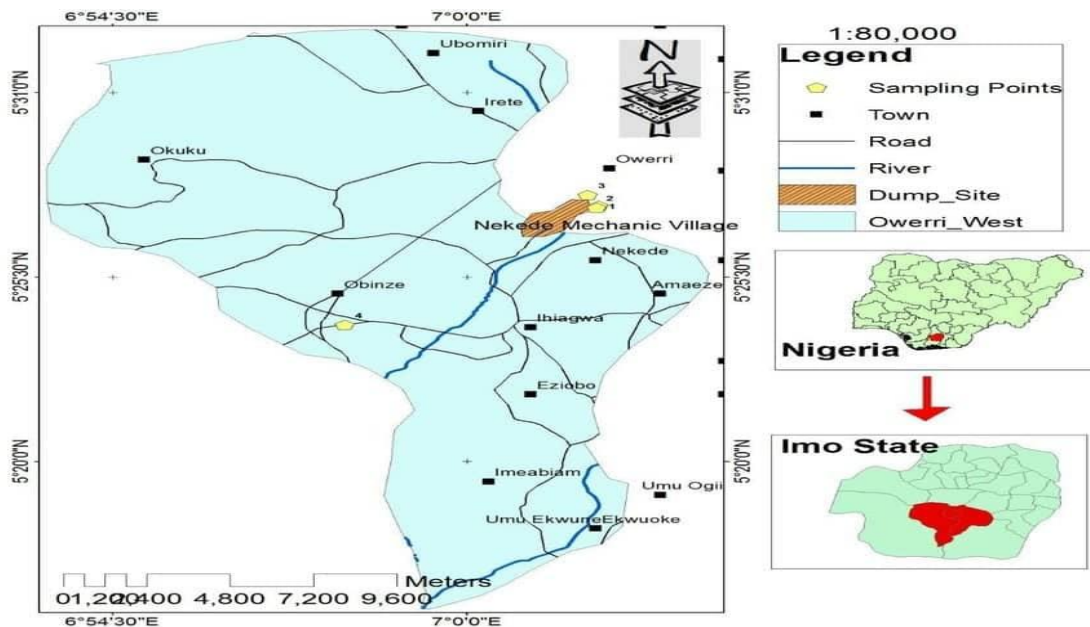


Figure 2: Map of Owerri West indicating Nekede (the study area) Source: (RECTAS, 2013)

Sample Collection

Soil samples were collected from three different locations in the test plot (experimental layout), and this was used in determining the various soil properties that influence the infiltration of water e.g permeability, hydraulic conductivity, moisture content etc, into it.

Different water samples were also collected from different sources in different locations. The water samples collected from different sources, whose viscosities, turbidity and densities were to be tested to determine their infiltration rate into the soil, include the following: Direct rainwater (uninterrupted); Surface water sample (from river; upstream and down stream); Borehole water (underground water); Hydraulic well water (underground water); Hand-dug well water (underground water); Earth pond water; Concrete pond water.

Materials

The following materials/apparatus were used for the soil laboratory analysis e.g for grain size determination; Soil auger borer, Oven, Evaporating dish, Rubber tubing connected to a water tap with a small spray, such as the rose from a watering can, Set of sieves arranged in a descending order, Weighing balance.

For the moisture content determination, materials/apparatus include: Weighing balance; Weighing containers / moisture cans; Desiccator; Oven set at 105⁰C – 110⁰C.

For hydraulic conductivity determination, materials/apparatus include: Permeameter (including falling head filter tank and manometer tubes); Ring stand with test tube clamp; Timing device (stop watch); Thermometer; Burette (100ml); Mixing pan; Vacuum pump; Compaction equipment.

For the infiltration analysis, materials/apparatus includes: Double ring infiltrometer; stopwatch; Meter rule; Wooden mallet; Container with a known volume of water; Marker.

Apparatus/materials used for the various water analysis includes, e.g for kinematic viscosity test: Water bath (H.H.S), U-tube viscometer, Retort stand, Syringe, Conical flask, Beaker, Stopwatch.

For the density determination, materials/apparatus includes: Specific bottle; Measuring cylinder; Electrically operated weighing balance; Neat and soft textile (used for cleaning).

For the turbidity determination, materials/apparatus includes: Nephelometer/turbidity meter; Sample cells; Standard flask; Funnel; Wash bottle; Glass beaker; Ionized water.

Methods

Topographic Investigation and Time of Concentration

The topographic investigation of the farm area (test plot), was done so as to provide useful information on the general lay of the farm area in order to ensure that the farm area is flat so as to avoid irregularities in the result of the infiltration analysis.

The contour map of the study area was extracted from the institution/school plan at a scale of 1:10,000. The length of flow of the farm area was measured to be 1180m from the plan, the elevation of the farthest point from the outlet is 46.5m, and the elevation of the outlet is 26.5m. With these, the elevation difference was estimated to be 20m, and the slope of the area was determined using the equation below

and the slope of the area was determined using the equation below:

$$\text{Slope} = \frac{\text{Difference in Elevation}}{\text{Maximum length of flow}} \quad \text{equ. 1}$$

$$\text{Slope} = \frac{\text{Elevation of the farthest point from the outlet} - \text{Elevation of the outlet}}{\text{Maximum length of flow}}$$

$$\frac{46.5 - 26.5 \text{ (m)}}{1180 \text{ (m)}}$$

$$= \frac{20}{1180} = 0.017 \text{ or } 1.7\%$$

∴ The slope of the farm area = 1.7%.

According to Gerick & Smithers (2014), the time of concentration of the test plot (farm area) was determined using the Kirpich empirical equation (Kirpich, 1940).

$$T_C = 0.0195L^{0.77}Sg^{-0.385} \quad \text{equ 2}$$

Where: T_C = the time of concentration (mins)

L = the maximum length of flow (m)

Sg = the watershed gradient in m/m or the difference in elevation between the outlet and the most remote point divided by the length L

Therefore:

$L = 1180\text{m}$ and

$$Sg = \frac{20\text{(m)}}{1180\text{m}}$$

$$T_C = 0.019 \times (1180)^{0.77} \times \left(\frac{20}{1180}\right)^{-0.385}$$

$$T_C = 0.019 \times 231.9258 \times 4.8059364$$

$$T_C = 4.4065902 \times 4.8059364$$

$$T_C = 21.18 \text{ min}$$

Therefore, the time of concentration (T_C) of the test plot = 21.18 min.

Soil Analysis

The different soil analyses conducted in the course of this research were necessary to determine the important physical properties and nature of the soil, which influence the downward movement of water from the soil surface into the soil (infiltration) (Brady & Weil, 2016). Among the properties tested for in the course of the analysis are the following: grain size analysis, hydraulic conductivity (permeability), available moisture content (MC) and the infiltration rate of the soil.

The wet sieving method was selected for the grain size determination, and this method was selected based on visual inspection of the texture of the soil, which was found to be fine-to medium-sized sand (Dane, & Topp, 2020). Soil samples were collected using the soil auger borer, from three different locations in the test plot, at a required depth. The locations were tagged as Location A, B, and C for proper identification.

The moisture content determination was done using the gravimetric method (oven drying method) on samples collected from the three different locations A, B and C, so as to ensure accurate determination of the initial moisture content of the soil from the test plot (ASTM, 2021).

Hydraulic conductivity is the rate of flow of water through the soil (Michael and Ojha, 2005). It greatly influences the infiltration possibilities of an area and depends on the properties of the soil and fluid. The falling head laboratory method was used to determine the hydraulic conductivity of the soil samples collected from Locations A, B, and C.

The infiltration test for this research work was done in-situ in the test plot using a double ring infiltrometer. It was done under a favourable climatic condition to ensure accuracy. The test was carried out to determine the velocity or speed at which water enters the soil from the surface and the depth in cm of water level that can enter the soil per hour is usually measured.

Water Analysis

The water analysis required for this work has to do with analytically experimenting on the selected physical qualities associated with water, which influence its downward movement into the soil (infiltration). The basic properties of water that was analyzed for this research work, as they influence infiltration rate into the soil, include the following: viscosity, density and turbidity. Since it is a known fact that these factors are temperature dependent, the various samples used were tested under controlled temperature ranges of 20⁰C, 25⁰C, 30⁰C and 35⁰C for each sample and factor as well. The temperature ranges were selected as they are in tandem with the atmospheric temperature during the irrigation period.

The kinematic viscosity for the various water samples was tested under a controlled temperature range of 20⁰C, 25⁰C, 30⁰C and 35⁰C using a U-tube viscometer.

The viscosity (kinematic) of the various water samples from different sources was calculated using the equation $V = C \times t$ *equ 3*

The density experiment for the various water samples was conducted under the required temperature ranges (i.e., 20, 25, 30, and 35⁰C).and their densities computed using the equation;

$$\text{Density} \rho = \frac{W_1 - W_0}{V} \quad \text{equ 4}$$

For turbidity determination, the analysis was done for the various water sample under the required controlled temperature range (i.e., 20, 25, 30, and 35⁰C) using the turbidimeter.

Statistical Analysis

A two-way analysis of variance (ANOVA) was performed for each response variable, with Water Source (12 levels) and Temperature (4 levels) as the two independent factors, to determine whether each variable had a statistically significant influence on each physical/quality variable. The Source $\times$ Temperature interaction parameter could not be separated from the residual error because each Source $\times$ Temperature combination was expressed by one measurement (i.e., no replicate measurements). The two main effects (Source and Temperature) were thus evaluated against a pooled residual (interaction/error) term with 33 degrees of freedom in a two-way ANOVA without replication (an additive, randomized-block-type model). The means were separated using Tukey's Honestly Significant Difference (HSD) post-hoc test when a main effect was significant ($p < 0.05$). Every test was performed at the significance level of 5% ($\alpha = 0.05$).The Shapiro-Wilk test was used to determine residual normality. Viscosity and density residuals were examined on their raw scale and showed a normal distribution ($p > 0.05$). As is customary for positively-skewed turbidity/concentration data, a log10 transformation was applied before the ANOVA and post-hoc comparisons because turbidity values breached the normality assumption on the raw scale (Shapiro–Wilk $p = 0.003$) and ranged over nearly three orders of magnitude (2–1437 NTU).

IV. Results

Study Area Parameters

Parameter	Result
Area of test plot	10,000 m ²
Maximum length of flow	1180 m
Elevation of farthest point	46.5 m
Elevation of outlet point	26.5 m
Slope of test area	1.7%
Time of concentration (T _c)	21.18 min

The climatic condition of the study area/test plot, shows that the area is sub-equatorial, with a high humidity of about 80–100%, and a moderate temperature of 25–35°C. The mean annual rainfall of the area is between 2000 mm and 2500 mm, with a rainfall intensity of 22 mm/hr.

Soil Analysis

Soil parameter	Location A	Location B	Location C
Average moisture content (%)	4.45	5.21	4.90
Hydraulic conductivity (cm/s)	6.505×10^{-3}	6.313×10^{-3}	6.505×10^{-3}

The average of the permeability test result produced from the three different locations in the experimental test plot was recorded as $K = 6.44 \times 10^{-3}$ cm/sec.

Particle-size result	Value
Percentage sand	81%
Percentage silt	18%
Average percentage of fines	17.7%
Soil dominant in the test plot area	loamy sand-loam

Infiltration Characteristics

Location	Initial infiltration rate (cm/h)	Final/steady infiltration rate (cm/h)	Elapsed time
A	15.6	7.8	150 min
B	25.2	6.6	150 min
C	20.4	7.8	150 min

The infiltration curves from the graphs plotted for the various water samples displays a similar trend in the behavior of water infiltration into the soil. At the beginning of the experiment the infiltration rates were initially high, but gradually decreases with time until it approached a nearly constant value.

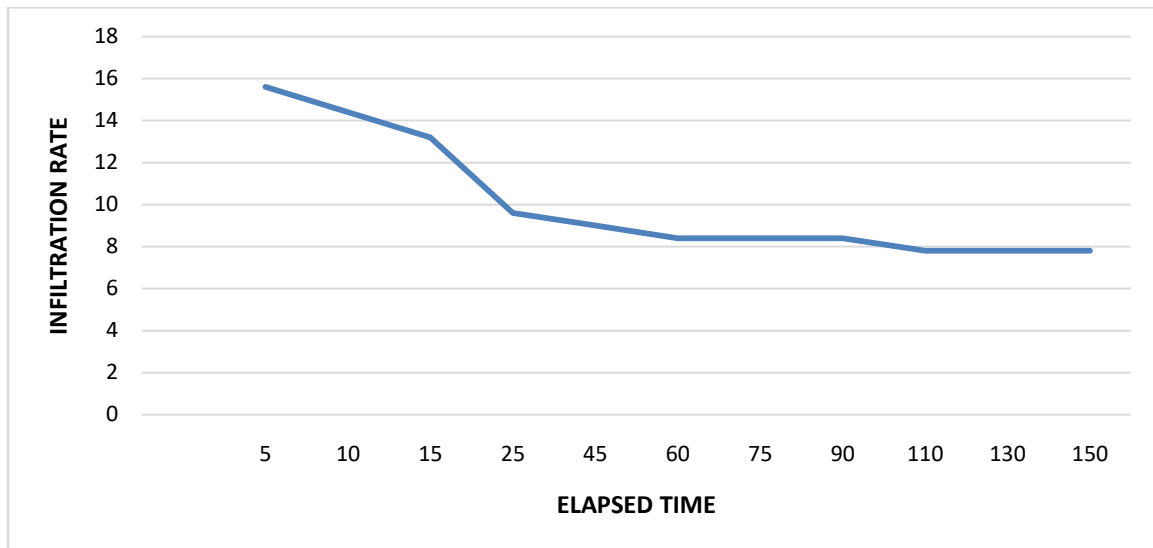


Figure 3: Infiltration Test Curve on Plot A before Water Analysis

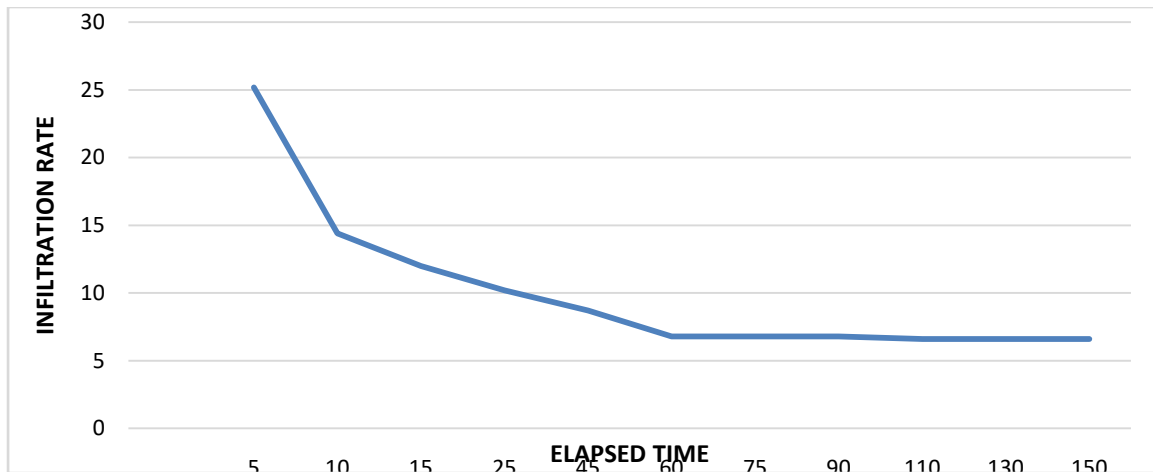


Figure 4: Infiltration Test Curve on Plot B before Water Analysis

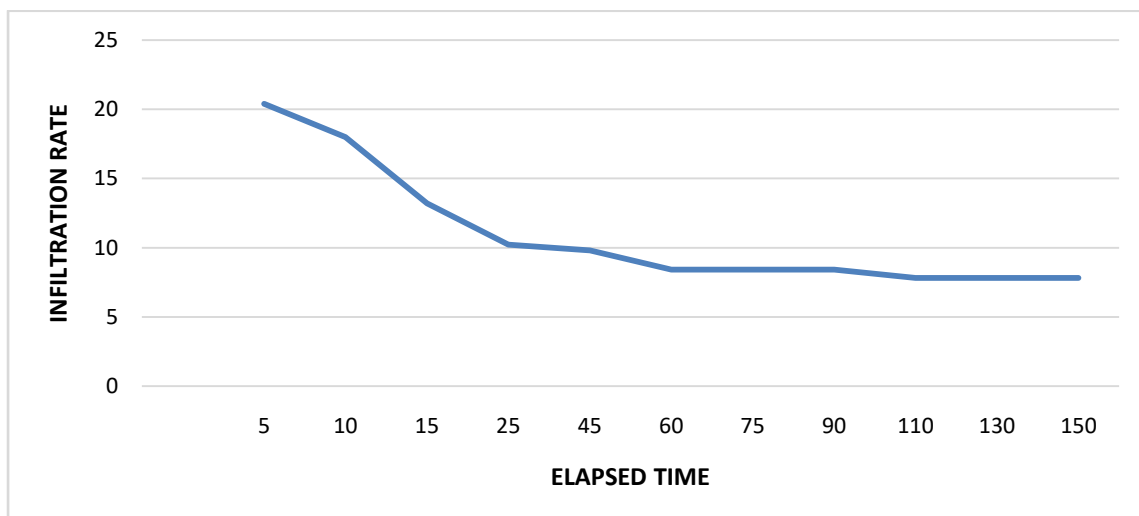


Figure 5: Infiltration Test on Plot C before Water Analysis

Kinematic Viscosity

Water source	20°C	25°C	30°C	35°C
Borehole water				
Owerri Municipal	5.08	4.59	4.55	4.36
Direct rainfall	4.57	4.41	4.32	4.29
Hydraulic well				
Ihiagwa	5.03	4.57	4.44	4.41
Hydraulic well				
Obinze	4.51	4.49	4.39	4.29
Well water				
Okigwe	4.60	4.57	4.52	4.36
Imo River up stream	5.20	4.54	4.36	4.27
Imo River downstream	5.40	4.57	4.43	4.35
Surface water				
Nekede	5.52	5.08	4.33	4.23
Concrete pond				
Nekede	5.02	4.59	4.47	4.29
Earth pond				
Agbala	5.03	4.57	4.40	4.22
Otamiri River upstream	5.40	4.48	4.30	4.20
Otamiri River downstream	5.50	4.55	4.46	4.30
ANOVA source	F	P	R²	
Water Source	1.313	0.261 (ns)	0.789	
Temperature	36.237	0.000**		
Temperature	Mean viscosity (mPa·s)		Tukey group	
20	5.072		a	
25	4.584		b	
30	4.414		bc	
35	4.298		c	

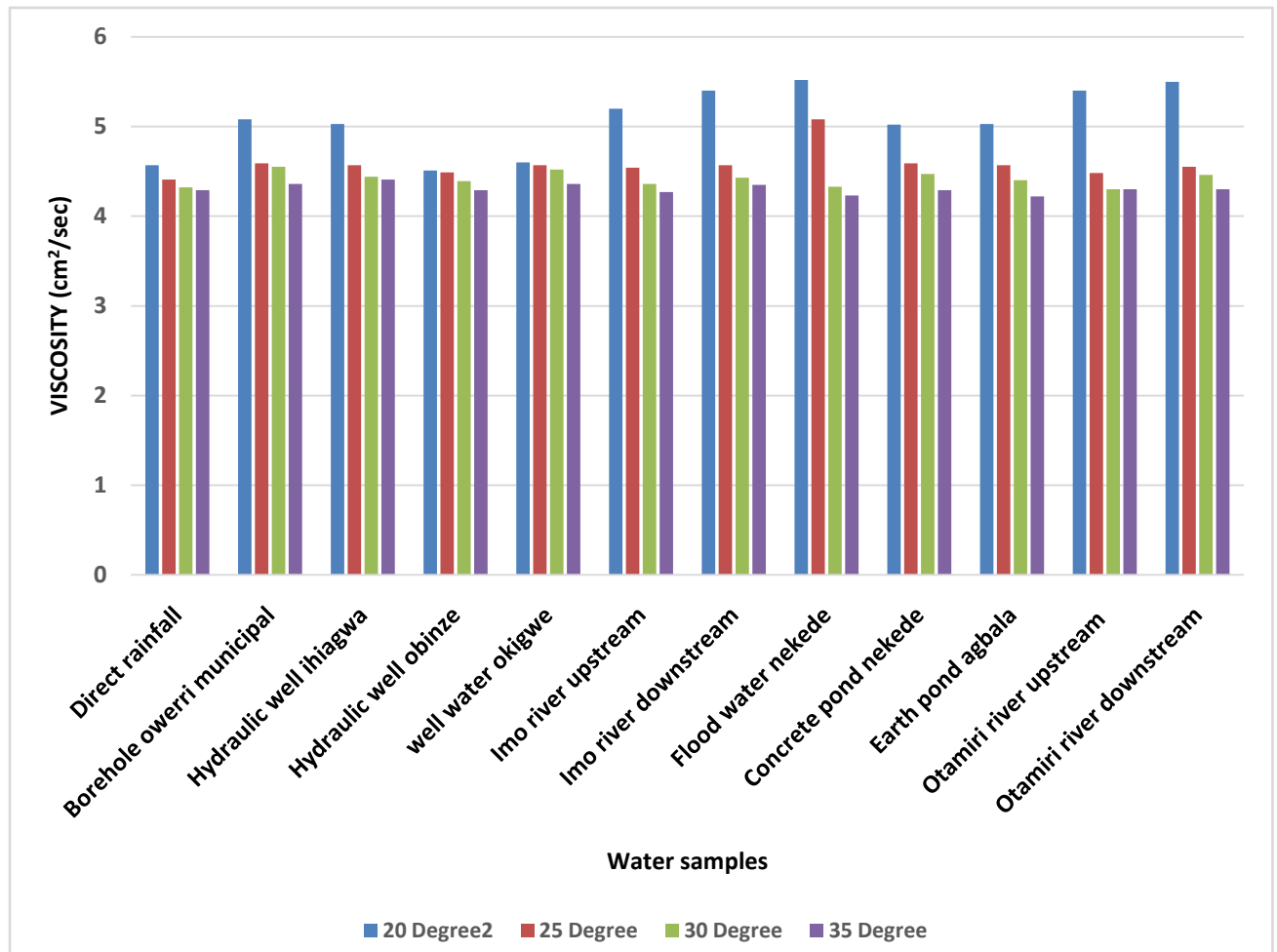


Figure 6: Water Viscosity Test Chart at Different Temperature Range

Density

Water source	20°C	25°C	30°C	35°C
Direct rainfall	0.9626	0.9622	0.9610	0.9592
Borehole water				
Owerri Municipal	0.9989	0.9970	0.9956	0.9922
Hydraulic well				
Ihiagwa	0.9462	0.9358	0.9264	0.9240
Hydraulic well				
Obinze	0.9614	0.9528	0.9382	0.9352
Well Water				
Okigwe	0.9672	0.9515	0.9434	0.9314
Imo River				
upstream	0.9626	0.9486	0.9366	0.9284
Imo River				
downstream	0.9712	0.9584	0.9448	0.9386
Surface water				
Nekede	0.9666	0.9598	0.9464	0.9286
Concrete pond				
Nekede	0.9672	0.9567	0.9483	0.9234
Earth pond				
Agbala	0.9810	0.9642	0.9481	0.9366
Otamiri River				
upstream	0.9620	0.9521	0.9491	0.9352
Otamiri River				
downstream	0.9616	0.9582	0.9492	0.9368
ANOVA source	F	P	R ²	
Water Source	24.475	0.000**	0.926	
Temperature	47.468	0.000**		
Temperature	Mean density (g/cm ³)		Tukey group	
20	0.9674		a	
25	0.9581		ab	
30	0.9489		bc	
35	0.9391		c	

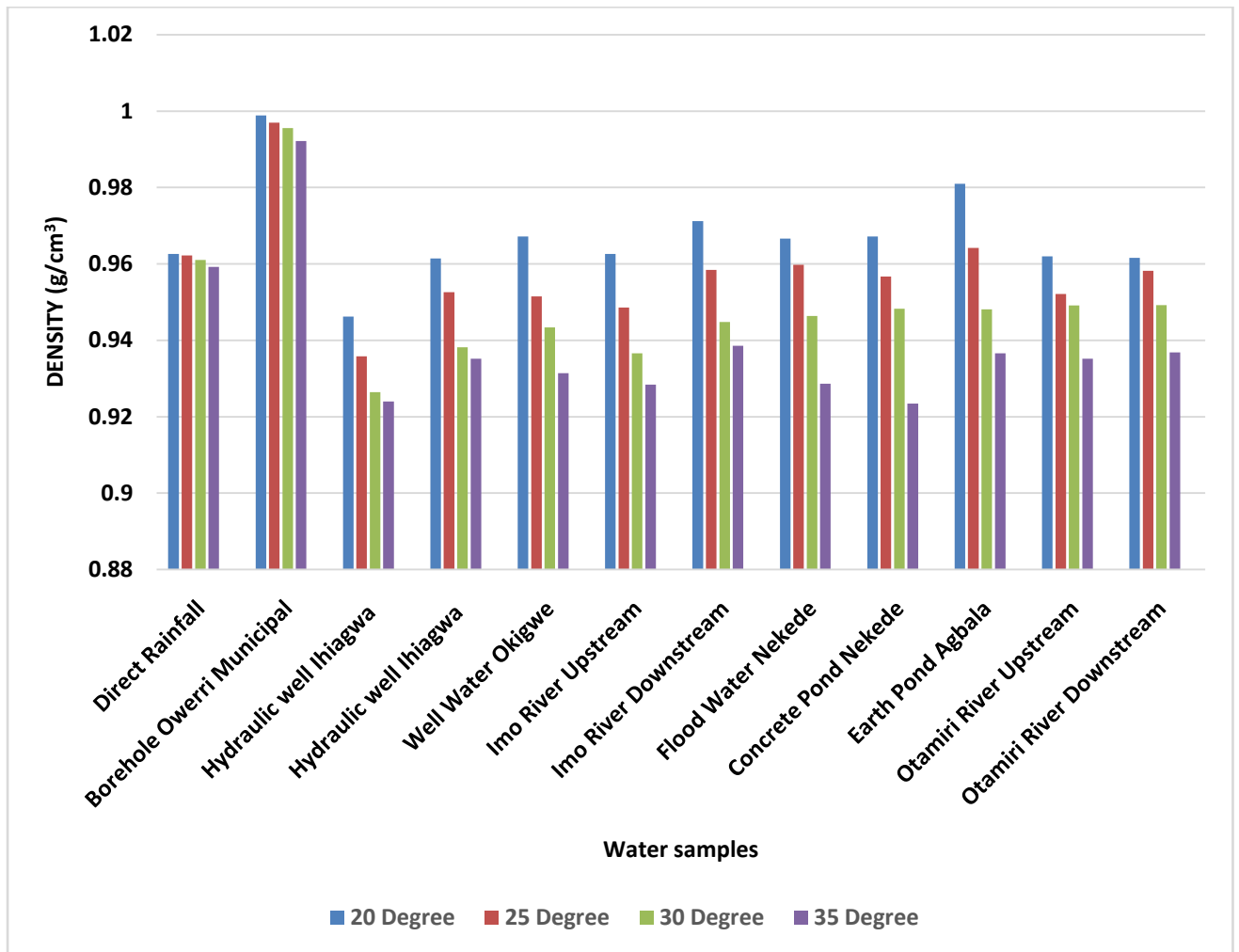


Figure 7: Water Density Test Chart at Different Temperature Range.

Turbidity

Water source	20°C	25°C	30°C	35°C
Direct rainfall	5.45	5.58	5.62	5.65
Borehole water				
Owerri Municipal	2.02	2.36	2.68	2.75
Hydraulic well				
Ihiagwa	10.34	12.45	15.37	17.60
Hydraulic well				
Obinze	4.65	4.82	4.96	5.02
Well Water				
Okigwe	4.46	4.62	4.84	5.08
Imo River				
upstream	176.4	182.8	191.6	209.8
Imo River				
downstream	241.0	245.0	249.0	257.0
Surface water				
Nekede	530.0	540.0	550.0	560.0
Concrete pond				
Nekede	915.0	920.0	925.0	930.0
Earth pond				
Agbala	1413.0	1420.0	1425.0	1437.0
Otamiri River				
upstream	40.0	42.0	45.0	48.0
Otamiri River				
downstream	60.0	62.0	65.0	68.0
ANOVA source	F		P	
Water Source	4356.1		< 0.001	
Temperature	17.27		< 0.001	

Mean turbidity increased from 270.3 NTU at 20°C to 297.9 NTU at 35°C. The lowest turbidity was recorded for borehole water (2.45 NTU), while other groundwater and rainfall sources ranged from 4.75–13.94 NTU. The open-flow river sources, including Otamiri and Imo Rivers, recorded higher turbidity values ranging from 40–241 NTU, while standing-water bodies showed the highest values, including Surface water Nekede (530 NTU), Concrete Pond Nekede (915 NTU), and Earth pond Agbala (1413 NTU).

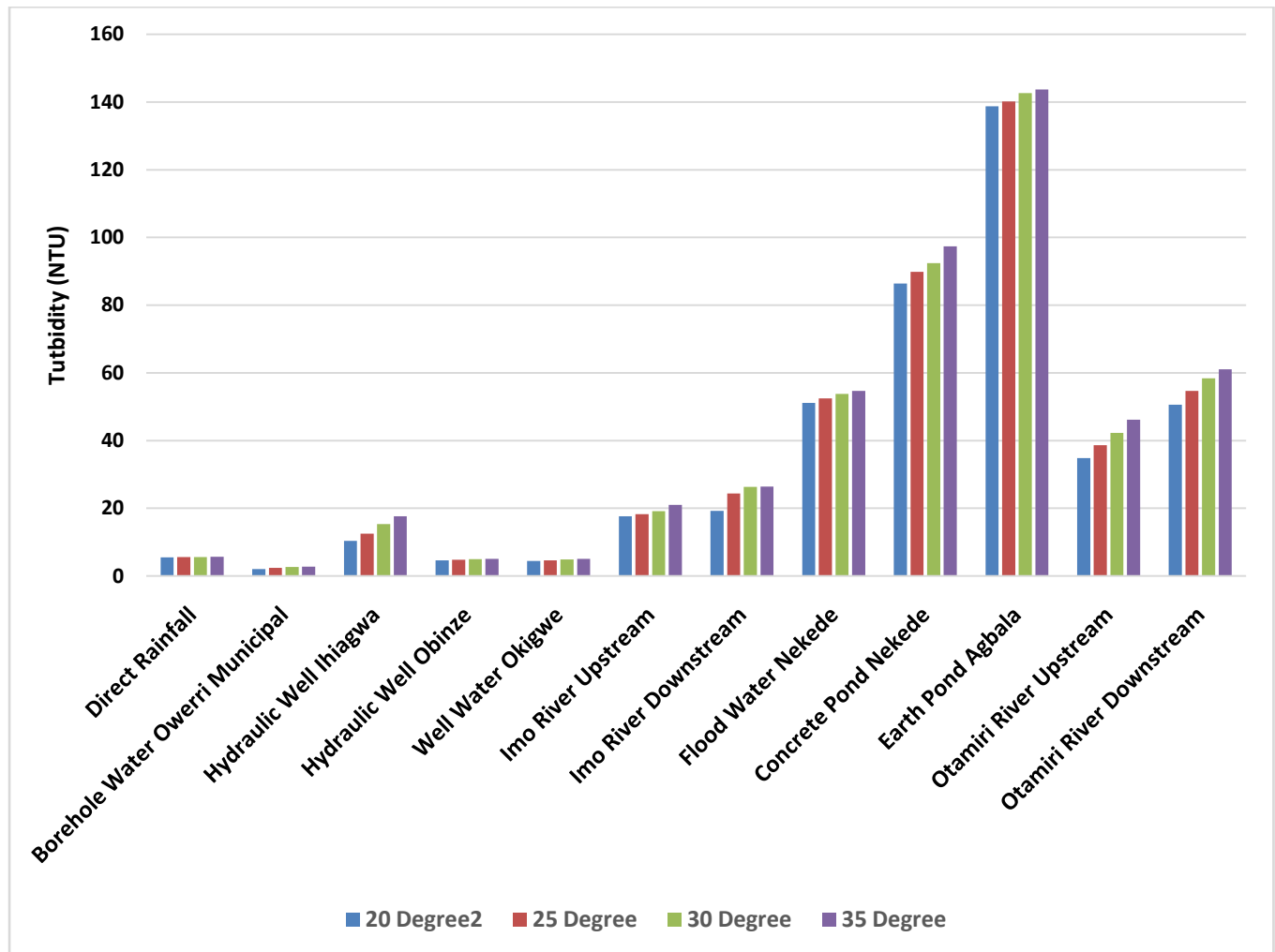


Figure 8: Water Turbidity Test Chart at Different Temperature Range.

Infiltration Trials after Water Analysis

Water sample	Initial infiltration rate (cm/h)	Rate at 60 min (cm/h)
Direct rainfall	16.20	7.60
Borehole water, Owerri Municipal	16.70	7.59
Hydraulic well, Ihiagwa	17.00	7.65
Hydraulic well, Obinze	17.50	7.78
Well water, Okigwe	16.50	7.65
Imo River upstream	16.70	7.50
Imo River downstream	17.30	7.50
Surface water, Nekede	20.20	7.50
Concrete pond, Nekede	19.40	7.50
Earth pond, Agbala	19.60	7.70
Otamiri River upstream	17.10	7.50
Otamiri River downstream	17.50	7.50

Discussions

The topographic survey conducted showed that the plot has a total area of 10,000m² with a maximum flow length of 1180m. Also, the elevation of the farthest point was determined to be 46.5m, while the elevation of the outlet point was 26.5m, which resulted in a total elevation difference of 20m across the test plot. From the values obtained, the average slope of the plot was calculated to be 1.7%. The calculated slope value shows that the experimental plot has a gentle slope/gradient, which conforms to the classification of a relatively flat terrain as reported by Schwab et al. (2021).

The climatic characteristics of the study area show that the area lies within the sub-equatorial climatic zone, which is typical of many regions in southern Nigeria. The recorded relative humidity ranges between 80%-100%, while the temperature ranges between 25⁰C and 35⁰C. Also, it was observed that the area experiences a mean annual rainfall of between 2000mm and 2500mm, with a rainfall intensity of approximately 22mm/hr.

The calculated time of concentration (T_C) of 21.18 minutes suggests that the experimental plot has a relatively rapid hydrologic response. Therefore the (T_C) is suitable for infiltration experiments because it allows sufficient observation of the initial and transitional stages of infiltration.

The moisture content analysis conducted on the soil samples from the three different locations (A, B, and C) in the study area, showed the average moisture as 4.45%, 5.21% and 4.90%, respectively. These results show that the soil within the experimental plot area possesses relatively low moisture content and are fairly uniform.

The wet sieve analysis conducted on the soil samples collected from the three different locations showed that the soil comprised an average of approximately 81% sand and 18% fines. Based on the USDA soil textural classification system, this particle size distribution classifies the soil as loamy sand. This textural composition provides the soil with relatively high permeability while retaining sufficient fine particles to impart moderate cohesion and limited water-holding capacity.

The hydraulic conductivity results obtained at the three sampling locations are nearly identical, with only negligible differences among Locations A, B, and C. This close agreement indicates that the soil within the experimental plot is relatively homogeneous with respect to its hydraulic properties. Such uniformity is advantageous because it minimizes spatial variability during infiltration experiments and ensures that differences observed in infiltration behaviour can be attributed primarily to the experimental variables—namely water viscosity, density, turbidity, and temperature rather than to inherent variations in soil properties.

The infiltration tests conducted at the three sampling locations within the experimental plot showed a similar infiltration pattern despite differences in their initial infiltration rates. At Location A, the infiltration rate decreased from 15.6 cm h⁻¹ to approximately 7.8 cm h⁻¹ after 150 minutes. Similarly, Location B recorded the highest initial infiltration rate of 25.2 cm h⁻¹, which gradually declined to 6.6 cm h⁻¹, while Location C exhibited an initial infiltration rate of 20.4 cm h⁻¹, eventually stabilizing at approximately 7.8 cm h⁻¹.

The observed decline in infiltration rate with time followed by the attainment of a relatively constant value is characteristic of infiltration processes in natural soils. The infiltration behaviour observed in this study agrees with the infiltration theory proposed by Horton (1940), who explained that infiltration capacity is initially high at the beginning of rainfall or water application but decreases progressively as the soil becomes wetter until a constant or equilibrium infiltration rate is attained.

The results obtained for the kinematic viscosity of the various water samples showed a consistent decrease in viscosity with increasing temperature. Similar trends were observed for all the water samples analysed, indicating that increasing temperature reduces the resistance of water to flow regardless of its source. The statistical analysis further confirmed that viscosity was significantly influenced by temperature ($F(3,33) = 36.24$, $p < 0.001$), but not by water source ($F(11,33) = 1.31$, $p = 0.261$).

The mean viscosity decreased progressively from 5.072 mPa·s at 20°C to 4.298 mPa·s at 35°C, representing an overall reduction of approximately 15.3% across the 15°C temperature range. The lower viscosity values obtained at higher temperatures imply that water would experience less resistance while moving through the interconnected pore spaces of the loamy sand soil used in the experiment.

The density measurements obtained for the various water samples showed a slight decrease with increasing temperature. For example, the density of the borehole water sample collected from Owerri Municipal decreased from 0.9989 g cm⁻³ at 20°C to 0.9922 g cm⁻³ at 35°C. The statistical analysis further showed that both temperature ($F(3,33) = 47.47$, $p < 0.001$) and water source ($F(11,33) = 24.48$, $p < 0.001$) significantly influenced water density. The model explained 92.6% of the total variation in density ($R^2 = 0.926$).

Although density contributes to the gravitational driving force responsible for water movement through soil, the variations observed between 20°C and 35°C were relatively small. Consequently, these density differences are unlikely to produce a significant independent effect on the movement of water through soil pores or on infiltration under normal irrigation conditions.

The turbidity measurements obtained for the various water samples revealed considerable differences in water clarity among the sources. Borehole water and direct rainfall recorded relatively low turbidity values, indicating comparatively clear water with low concentrations of suspended particles. In contrast, surface water sources such as the Imo River, Otamiri River, floodwater, and earth pond water exhibited very high turbidity values, with some samples exceeding 1400 NTU.

The statistical analysis showed that water source contributed the largest portion of the variance in turbidity ($F(11,33) = 4356.1$, $p < 0.001$), while temperature also had a statistically significant but very small influence ($F(3,33) = 17.27$, $p < 0.001$). This demonstrates that the physical source and exposure of the water body, rather than the temperature at which it was analysed, was the primary factor influencing turbidity.

The mean turbidity increased moderately with temperature, from 270.3 NTU at 20°C to 297.9 NTU at 35°C. Although temperature had a statistically significant effect, none of the pairwise temperature comparisons remained significant after Tukey correction, confirming that the temperature effect was minor compared with the influence of water source.

From the perspective of soil infiltration, the results indicate that water samples with high turbidity possess greater potential to reduce infiltration than clearer water. During irrigation or infiltration, suspended particles may accumulate on the soil surface and within pore spaces, gradually forming a thin sealing or clogging layer that restricts water movement into the soil. Consequently, infiltration rates and saturated hydraulic conductivity decrease as pore continuity is progressively reduced.

The findings therefore reveal that the three water quality parameters do not contribute equally to infiltration under varying thermal conditions. Among the parameters investigated, viscosity exhibited the greatest sensitivity to temperature, density showed only minor temperature-dependent changes, whereas turbidity exerted its influence largely through sediment-induced pore clogging independent of temperature.

Accordingly, the present study refines the conclusions of Arora (2009), Hillel (2013), and Michael (2018) by demonstrating that water quality parameters should not be evaluated independently of temperature when assessing infiltration behaviour. Rather than considering viscosity, density, and turbidity as isolated variables, the present research shows that temperature modifies the behaviour of these properties—particularly viscosity—and consequently influences the infiltration process.

V. Conclusion and Recommendations

Conclusion

This study successfully evaluated the effects of selected water quality parameters under a controlled temperature range on soil infiltration characteristics, using viscosity, density, turbidity, and infiltration analysis on various water samples obtained from rivers, ponds, boreholes, wells, rainfall, and surface water sources.

The experimental results revealed that viscosity and density decreased with increasing temperature for all water samples, while turbidity generally increased with temperature. The three two-way ANOVAs revealed a physically cohesive and uniform trend. Viscosity was mostly controlled by temperature and unaffected by the source of water, confirming that it is an inherent thermophysical characteristic of water. Temperature and water source had a substantial impact on density, reflecting the combined effects of thermal expansion and dissolved mineral/solute loading. Temperature had only a small, secondary impact on turbidity, while standing or pond waters were far more turbid than groundwater.

The infiltration tests showed that infiltration rates were initially high and gradually decreased with time until a constant infiltration rate was attained. This behaviour was due to gradual soil saturation and a reduction in soil suction forces. Water samples with low turbidity, such as direct rainfall and borehole water, produced more stable infiltration characteristics because they contained fewer suspended particles capable of clogging soil pores. Highly turbid water samples such as earth pond water, concrete pond water, and surface water from Nekede contained large amounts of suspended solids, which contributed to pore blockage and reduced infiltration efficiency during the later stages of infiltration.

The study further establishes the fact that among the selected water quality parameters, turbidity significantly influences infiltration processes and hydraulic behaviour of soils, while temperature variation also played an important role by altering viscosity and density characteristics of the water samples. Among the parameters investigated, viscosity was the most temperature-sensitive, density exhibited only minor temperature-induced variations, and turbidity influenced infiltration primarily through sediment deposition rather than thermal effects.

The findings show that even though the twelve water samples were taken from the same common area, their physicochemical characteristics varied significantly depending on the hydrological origin. They also show that temperature is an important control factor whenever viscosity or density measurements are compared across studies or treatment processes like filtration, sedimentation, and pumping design, where both variables directly affect hydraulic performance and flow behaviour.

Conclusively, giving much focus to the turbidity nature of water samples, and minor considerations to viscosity and density, cleaner water sources are more suitable for infiltration and groundwater recharge processes, as highly turbid water may gradually reduce soil permeability and infiltration efficiency over time during irrigation processes.

Recommendation

1. Particular attention should be given to turbidity and temperature because these parameters significantly influence soil infiltration and irrigation efficiency.
2. Where practicable, infiltration/irrigation should be carried out during periods when water temperatures favour lower viscosity.
3. Farmers and irrigation managers should incorporate water quality monitoring into irrigation management programmes.
4. Water intended for drip and sprinkler irrigation systems should possess low turbidity levels. This will reduce emitter clogging, improve water distribution uniformity, minimize maintenance costs, and enhance irrigation system performance.
5. Government agencies, irrigation authorities, and water resource managers should develop practical guidelines for water quality that include physical parameters such as viscosity, density, and turbidity, in addition to conventional chemical quality standards.
6. Agricultural extension services should educate farmers on the influence of water quality on soil infiltration and irrigation efficiency.
7. Further studies should evaluate a wider temperature range and additional water quality parameters.
8. Mathematical and numerical models capable of predicting soil infiltration from measurable water quality parameters should be developed and validated.

The findings of this study should be considered in the design and management of irrigation schemes.

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